

MATH 535: LINEAR ALGEBRA

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1. AUGUST 24, 2026

1.1. Fields.

Definition 1 (Binary operation). *A binary operation on a set S is a function*

$$F: S \times S \longrightarrow S.$$

Thus, for every $x, y \in S$, the value $F(x, y)$ is again an element of S . We usually write $x \cdot y$ or $x + y$ in place of $F(x, y)$.

Example 1. *Addition on $\mathbb{Z}$ is a binary operation: the map*

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n$$

takes every pair of integers to an integer.

Definition 2. *Let F be a binary operation on S . A subset $T \subseteq S$ is closed under F if*

$$F(x, y) \in T$$

whenever $x, y \in T$.

Example 2. *The positive integers $\mathbb{Z}_{>0}$ are closed under addition. The odd integers are not closed under addition, but they are closed under multiplication.*

Definition 3 (Field). *Let $\mathbb{F}$ be a set equipped with two binary operations, called addition and multiplication. The sum and product of $a, b \in \mathbb{F}$ are denoted by $a + b$ and ab , respectively. Suppose that for every $a, b, c \in \mathbb{F}$,*

- (i) $a + b = b + a$;
- (ii) $(a + b) + c = a + (b + c)$;
- (iii) $ab = ba$;
- (iv) $a(bc) = (ab)c$;
- (v) $a(b + c) = ab + ac$.

Suppose further that

(vi) $\mathbb{F}$ contains an additive identity 0 and a multiplicative identity 1 , so that

$$a + 0 = a \quad \text{and} \quad 1a = a$$

for every $a \in \mathbb{F}$;

(vii) $1 \neq 0$;

(viii) every $a \in \mathbb{F}$ has an additive inverse $-a \in \mathbb{F}$ such that $a + (-a) = 0$;

(ix) every nonzero $a \in \mathbb{F}$ has a multiplicative inverse $a^{-1} \in \mathbb{F}$ such that $aa^{-1} = 1$.

Then $\mathbb{F}$ is called a field.

Example 3. The integers $\mathbb{Z}$ do not form a field: for example, 2 has no multiplicative inverse in $\mathbb{Z}$. The sets $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, with their usual operations, are fields.

There are also finite fields. For example,

$$\mathbb{F}_2 = \{0, 1\} \quad \text{and} \quad \mathbb{F}_3 = \{0, 1, 2\},$$

where the elements are the possible remainders upon division by 2 and 3 , respectively. Addition and multiplication in $\mathbb{F}_3$ are given by the following tables:

$$\begin{array}{c|ccc} + & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 \end{array} \quad \begin{array}{c|ccc} \cdot & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 \\ 2 & 0 & 2 & 1 \end{array}.$$

More generally, if p is prime, then

$$\mathbb{F}_p = \{0, 1, \dots, p-1\},$$

with addition and multiplication performed modulo p , is a field.

Lemma 1. If a is an element of a field $\mathbb{F}$, then $a \cdot 0 = 0$.

Proof. Since $0 + 0 = 0$, distributivity gives

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding the additive inverse of $a \cdot 0$ to both sides yields $0 = a \cdot 0$. $\square$

2. AUGUST 26, 2026

2.1. Further properties of fields. The preceding lecture showed that $a \cdot 0 = 0$ for every element a of a field. We now obtain the zero-product property.

Proposition 1 (Zero-product property). *Let $a, b \in \mathbb{F}$. If $ab = 0$, then either $a = 0$ or $b = 0$.*

Proof. Suppose that $ab = 0$ and $a \neq 0$. Since $\mathbb{F}$ is a field, a^{-1} exists. Therefore

$$0 = a^{-1}0 = a^{-1}(ab) = (a^{-1}a)b = 1b = b.$$

Thus $a = 0$ or $b = 0$. $\square$

2.2. Characteristic of a field. For a positive integer n and $a \in \mathbb{F}$, write

$$na = \underbrace{a + a + \cdots + a}_{n \text{ terms}}.$$

Definition 4 (Characteristic). *Let $\mathbb{F}$ be a field. If $n1 = 0$ for some positive integer n , then the least such n is called the characteristic of $\mathbb{F}$, denoted $\text{char}(\mathbb{F})$. If $n1 \neq 0$ for every positive integer n , then $\mathbb{F}$ has characteristic zero.*

The fields $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ have characteristic zero. For every prime p , the field $\mathbb{F}_p$ has characteristic p . The rational function field $\mathbb{F}_p(t)$ is an infinite field of characteristic p .

Proposition 2. *If $\mathbb{F}$ has positive characteristic r , then r is prime.*

Proof. We have $r \neq 1$, since $1 \neq 0$. Suppose that r is composite, so

$$r = st$$

for integers s, t satisfying $1 < s, t < r$. Distributivity gives

$$(s1)(t1) = (st)1 = r1 = 0.$$

By the zero-product property, either $s1 = 0$ or $t1 = 0$. Both alternatives contradict the minimality of r . Hence r is prime. $\square$

Proposition 3. *If $\text{char}(\mathbb{F}) = p > 0$, then*

$$pa = 0$$

for every $a \in \mathbb{F}$.

Proof. Since $p1 = 0$, associativity and distributivity imply

$$pa = p(1a) = (p1)a = 0a = 0.$$

□

Proposition 4. *Every finite field has positive characteristic.*

Proof. Let $\mathbb{F}$ be finite. The sequence

$$1, \quad 2 \cdot 1, \quad 3 \cdot 1, \quad \dots$$

cannot consist of distinct elements. Hence there are positive integers $m > n$ such that

$$m1 = n1.$$

Adding the additive inverse of $n1$ to both sides gives

$$(m - n)1 = 0.$$

Thus $\mathbb{F}$ has positive characteristic. □

2.3. Algebraically closed fields.

Definition 5 (Algebraically closed field). *A field $\mathbb{F}$ is algebraically closed if every nonconstant polynomial $f \in \mathbb{F}[x]$ has a root in $\mathbb{F}$.*

The field $\mathbb{Q}$ is not algebraically closed, because $x^2 - 2$ has no root in $\mathbb{Q}$. The field $\mathbb{R}$ is not algebraically closed, because $x^2 + 1$ has no real root. By the fundamental theorem of algebra, $\mathbb{C}$ is algebraically closed.

Proposition 5. *No finite field is algebraically closed.*

Proof. Let

$$\mathbb{F} = \{a_1, \dots, a_q\}$$

be finite. Consider the nonconstant polynomial

$$f(x) = 1 + \prod_{i=1}^q (x - a_i).$$

For each $i = 1, \dots, q$,

$$f(a_i) = 1 \neq 0.$$

Thus f has no root in $\mathbb{F}$, so $\mathbb{F}$ is not algebraically closed. $\square$

2.4. Vectors and vector spaces. The spaces $\mathbb{R}^2$, $\mathbb{R}^3$, and, more generally,

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\}$$

are the motivating examples of vector spaces.

Definition 6 (Vector space). *Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set V equipped with two operations,*

$$V \times V \longrightarrow V, \quad (u, v) \longmapsto u + v,$$

and

$$\mathbb{F} \times V \longrightarrow V, \quad (\alpha, v) \longmapsto \alpha v,$$

satisfying the following axioms for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{F}$:

- (V1) $u + (v + w) = (u + v) + w$;
- (V2) $u + v = v + u$;
- (V3) there is an element $0 \in V$ such that $v + 0 = 0 + v = v$;
- (V4) for every $v \in V$, there is an element $-v \in V$ such that $v + (-v) = (-v) + v = 0$;
- (V5) $1v = v$;
- (V6) $(\alpha\beta)v = \alpha(\beta v)$;
- (V7) $\alpha(u + v) = \alpha u + \alpha v$;
- (V8) $(\alpha + \beta)v = \alpha v + \beta v$.

The elements of V are called *vectors*, and the elements of $\mathbb{F}$ are called *scalars*.

3. AUGUST 28, 2026

3.1. Examples of vector spaces.

Example 4 (Coordinate spaces). *Let $\mathbb{F}$ be a field. The set*

$$\mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{F} \right\}$$

is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \alpha \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix}.$$

In particular, $\mathbb{R}^n$ is a vector space over $\mathbb{R}$ and $\mathbb{C}^n$ is a vector space over $\mathbb{C}$.

Example 5 (Matrix spaces). *The set $M_{m \times n}(\mathbb{F})$ of $m \times n$ matrices with entries in $\mathbb{F}$ is a vector space over $\mathbb{F}$. Addition and scalar multiplication are defined entrywise:*

$$[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}], \quad \alpha[a_{ij}] = [\alpha a_{ij}].$$

Example 6 (Polynomial spaces). *Let*

$$\mathcal{P}_n(\mathbb{F}) = \{a_0 + a_1t + \dots + a_nt^n : a_0, \dots, a_n \in \mathbb{F}\}.$$

This is the vector space over $\mathbb{F}$ of polynomials of degree at most n . Its operations are

$$\begin{aligned} (a_0 + a_1t + \dots + a_nt^n) + (b_0 + b_1t + \dots + b_nt^n) \\ = (a_0 + b_0) + (a_1 + b_1)t + \dots + (a_n + b_n)t^n, \end{aligned}$$

and

$$\alpha(a_0 + a_1t + \dots + a_nt^n) = \alpha a_0 + \alpha a_1t + \dots + \alpha a_nt^n.$$

3.2. Linear combinations and spanning sets.

Definition 7 (Linear combination and span). *Let V be a vector space over $\mathbb{F}$ and let $v_1, \dots, v_p \in V$. A linear combination of these vectors is a vector of the form*

$$\alpha_1 v_1 + \dots + \alpha_p v_p, \quad \alpha_1, \dots, \alpha_p \in \mathbb{F}.$$

The set of all their linear combinations is called their span:

$$\text{span}\{v_1, \dots, v_p\} = \{\alpha_1 v_1 + \dots + \alpha_p v_p : \alpha_1, \dots, \alpha_p \in \mathbb{F}\}.$$

A linear combination is trivial if all of its coefficients are zero.

In $\mathbb{R}^2$, let

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$\text{span}\{e_1, e_2\} = \mathbb{R}^2,$$

because

$$\begin{bmatrix} x \\ y \end{bmatrix} = x e_1 + y e_2.$$

If

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

then $\{e_1, e_2, v\}$ also spans $\mathbb{R}^2$, since

$$\begin{bmatrix} x \\ y \end{bmatrix} = x e_1 + y e_2 + 0 v.$$

The additional vector is unnecessary, which motivates the question of redundancy in a spanning set.

Definition 8 (Spanning set). *A collection $\{v_1, \dots, v_p\} \subseteq V$ is a spanning set for V if every $v \in V$ can be written as a linear combination*

$$v = \alpha_1 v_1 + \dots + \alpha_p v_p.$$

Equivalently,

$$\text{span}\{v_1, \dots, v_p\} = V.$$

3.3. Linear independence.

Definition 9 (Linear independence). *A collection $\{v_1, \dots, v_p\} \subseteq V$ is linearly independent if*

$$\alpha_1 v_1 + \dots + \alpha_p v_p = 0$$

implies

$$\alpha_1 = \cdots = \alpha_p = 0.$$

In other words, only the trivial linear combination of $v_1, \dots, v_p$ equals zero. A collection that is not linearly independent is called linearly dependent.

Example 7. Consider the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^3$. To test linear independence, solve

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

This gives the system

$$\begin{cases} \alpha + 2\beta + 3\gamma = 0, \\ \alpha + \gamma = 0, \\ \beta + \gamma = 0. \end{cases}$$

Taking $\gamma = 1$ gives $\alpha = \beta = -1$, so

$$-v_1 - v_2 + v_3 = 0.$$

Thus the vectors are linearly dependent; equivalently, $v_3 = v_1 + v_2$.

The following result precisely characterizes redundancy in a finite collection of vectors.

Proposition 6. Let $v_1, \dots, v_p \in V$. The collection $\{v_1, \dots, v_p\}$ is linearly dependent if and only if one of its vectors is a linear combination of the others. Equivalently, there is an index k and scalars $\beta_j \in \mathbb{F}$ such that

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j.$$

Proof. Suppose first that $\{v_1, \dots, v_p\}$ is linearly dependent. Then there are scalars $\alpha_1, \dots, \alpha_p$, not all zero, such that

$$\alpha_1 v_1 + \dots + \alpha_p v_p = 0.$$

Choose k such that $\alpha_k \neq 0$. Since $\mathbb{F}$ is a field, α_k^{-1} exists, and we can solve for v_k :

$$v_k = - \sum_{\substack{j=1 \\ j \neq k}}^p \alpha_k^{-1} \alpha_j v_j.$$

Thus v_k is a linear combination of the other vectors.

Conversely, suppose that for some k ,

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j.$$

Moving all terms to one side gives

$$v_k - \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j = 0.$$

This is a nontrivial linear combination of $v_1, \dots, v_p$, because the coefficient of v_k is $1 \neq 0$. Hence the collection is linearly dependent. $\square$

4. AUGUST 31, 2026

4.1. Bases and coordinates.

Definition 10 (Basis). *A collection of vectors $\{v_1, \dots, v_n\}$ in a vector space V is a basis for V if it is linearly independent and spans V .*

The two conditions in the definition can be combined into a statement about representing vectors uniquely.

Proposition 7. *A collection $\{v_1, \dots, v_n\}$ is a basis for V if and only if every vector $v \in V$ has a unique representation of the form*

$$v = a_1 v_1 + \dots + a_n v_n, \quad a_1, \dots, a_n \in \mathbb{F}.$$

When $\{v_1, \dots, v_n\}$ is a basis, the scalars $a_1, \dots, a_n$ are called the coordinates of v with respect to this basis.

Proof. Suppose first that $\{v_1, \dots, v_n\}$ is a basis. Since the collection spans V , every $v \in V$ has a representation

$$v = a_1v_1 + \cdots + a_nv_n.$$

To prove uniqueness, suppose that

$$v = \tilde{a}_1v_1 + \cdots + \tilde{a}_nv_n$$

is another such representation. Subtracting the two equations gives

$$(a_1 - \tilde{a}_1)v_1 + \cdots + (a_n - \tilde{a}_n)v_n = 0.$$

Because $\{v_1, \dots, v_n\}$ is linearly independent,

$$a_i - \tilde{a}_i = 0$$

for each i . Thus $a_i = \tilde{a}_i$ for every i , and the representation is unique.

Conversely, suppose every $v \in V$ has a unique representation as a linear combination of $v_1, \dots, v_n$. The existence of such a representation for every v shows that the collection spans V . Now suppose

$$a_1v_1 + \cdots + a_nv_n = 0.$$

The zero vector also has the representation

$$0 = 0v_1 + \cdots + 0v_n.$$

Uniqueness therefore gives $a_1 = \cdots = a_n = 0$. Hence the collection is linearly independent, so it is a basis for V . $\square$

Example 8 (The standard basis of $\mathbb{F}^n$). For $i = 1, \dots, n$, let $e_i \in \mathbb{F}^n$ be the vector whose i th coordinate is 1 and whose other coordinates are 0. Thus

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Every vector in $\mathbb{F}^n$ has the unique representation

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n.$$

Consequently, $\{e_1, \dots, e_n\}$ is a basis for $\mathbb{F}^n$, called its standard basis.

Example 9 (The standard basis of $\mathcal{P}_n(\mathbb{F})$). Recall that $\mathcal{P}_n(\mathbb{F})$ is the vector space of polynomials over $\mathbb{F}$ of degree at most n . Each polynomial has a unique expression

$$p(t) = a_0 + a_1 t + \cdots + a_n t^n.$$

Hence

$$\{1, t, t^2, \dots, t^n\}$$

is a basis for $\mathcal{P}_n(\mathbb{F})$, called its standard basis.

Proposition 8. Every finite nonempty spanning set for a vector space contains a basis.

Proof. Let $\{v_1, \dots, v_p\}$ span V . If this collection is linearly independent, then it is already a basis. Otherwise, by Proposition 6, some v_k is a linear combination of the other vectors. It follows that removing v_k does not change the span:

$$\text{span}\{v_1, \dots, v_p\} = \text{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_p\}.$$

If the remaining collection is linearly independent, it is a basis. If not, repeat the same argument. The original spanning set is finite, so this process must terminate. The final collection still spans V and is linearly independent, and is therefore a basis contained in the original spanning set. $\square$

4.2. Linear transformations. A *transformation*, also called a map or function, from a set X to a set Y is a rule $T: X \rightarrow Y$ that assigns to every $x \in X$ an element $T(x) \in Y$.

Definition 11 (Linear transformation). *Let V and W be vector spaces over the same field $\mathbb{F}$. A transformation $T: V \rightarrow W$ is linear if*

- (i) $T(u + v) = T(u) + T(v)$ for all $u, v \in V$;
- (ii) $T(\alpha u) = \alpha T(u)$ for all $u \in V$ and $\alpha \in \mathbb{F}$.

Equivalently, $T: V \rightarrow W$ is linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all $u, v \in V$ and $\alpha, \beta \in \mathbb{F}$.

Example 10 (Reflection). *Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by*

$$T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}.$$

Geometrically, T reflects each point across the first coordinate axis. To check additivity, observe that

$$\begin{aligned} T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right) &= T \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + y_1 \\ -x_2 - y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ -y_2 \end{bmatrix} \\ &= T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + T \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \end{aligned}$$

For $\alpha \in \mathbb{R}$,

$$\begin{aligned} T \left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= T \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha x_1 \\ -\alpha x_2 \end{bmatrix} \\ &= \alpha \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} = \alpha T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned}$$

Thus T is linear.

Example 11 (The differentiation operator). Let $n \geq 1$, $V = \mathcal{P}_n(\mathbb{F})$, and $W = \mathcal{P}_{n-1}(\mathbb{F})$. Define

$$T: V \longrightarrow W, \quad T(p) = p'.$$

Since

$$(p + q)' = p' + q' \quad \text{and} \quad (\alpha p)' = \alpha p',$$

the differentiation operator T is linear.

5. SEPTEMBER 2, 2026

5.1. Linear transformations determined by a basis. The following result shows that prescribing the values of a linear transformation on a basis completely determines the transformation.

Lemma 2 (Linear map lemma). Suppose that $\{v_1, \dots, v_n\}$ is a basis for V , and let $w_1, \dots, w_n$ be arbitrary vectors in W . Then there exists a unique linear transformation $T: V \rightarrow W$ such that

$$T(v_k) = w_k$$

for each $k = 1, \dots, n$.

Proof. Every $v \in V$ has a unique representation

$$v = \sum_{i=1}^n c_i v_i.$$

Define

$$T(v) = \sum_{i=1}^n c_i w_i.$$

The uniqueness of the coordinates $c_1, \dots, c_n$ shows that this definition is well-defined. In particular, taking $v = v_k$ gives $T(v_k) = w_k$.

To prove that T is linear, write

$$u = \sum_{i=1}^n a_i v_i \quad \text{and} \quad v = \sum_{i=1}^n c_i v_i.$$

Then

$$\begin{aligned}
 T(u + v) &= T\left(\sum_{i=1}^n (a_i + c_i)v_i\right) \\
 &= \sum_{i=1}^n (a_i + c_i)w_i \\
 &= \sum_{i=1}^n a_i w_i + \sum_{i=1}^n c_i w_i \\
 &= T(u) + T(v).
 \end{aligned}$$

Similarly, for every $\alpha \in \mathbb{F}$,

$$\begin{aligned}
 T(\alpha v) &= T\left(\sum_{i=1}^n \alpha c_i v_i\right) \\
 &= \sum_{i=1}^n \alpha c_i w_i = \alpha \sum_{i=1}^n c_i w_i = \alpha T(v).
 \end{aligned}$$

Thus T is linear.

It remains to prove uniqueness. Suppose that $T': V \rightarrow W$ is another linear transformation satisfying $T'(v_k) = w_k$ for every k . If $v = \sum_{i=1}^n c_i v_i$, then

$$T'(v) = \sum_{i=1}^n c_i T'(v_i) = \sum_{i=1}^n c_i w_i = T(v).$$

Hence $T' = T$. □

5.2. Matrix representations of linear transformations. An *ordered basis* is a basis whose vectors are listed in a fixed order. Let

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m)$$

be ordered bases for V and W , respectively. For $v = x_1 v_1 + \dots + x_n v_n$, write

$$[v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

for the coordinate vector of v with respect to $\mathcal{B}$.

Theorem 1 (Matrix representation theorem). *Let V and W be vector spaces over $\mathbb{F}$ with ordered bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$. For every linear transformation $T: V \rightarrow W$, there is a unique matrix*

$$A = [A_{ij}] = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \in M_{m \times n}(\mathbb{F})$$

such that

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i$$

for each $j = 1, \dots, n$. Conversely, every matrix $A \in M_{m \times n}(\mathbb{F})$ determines a unique linear transformation $T: V \rightarrow W$ through this formula.

Proof. Let $T: V \rightarrow W$ be linear. Because $\mathcal{C}$ is a basis for W , each vector $T(v_j)$ has a unique representation

$$T(v_j) = A_{1j}w_1 + \cdots + A_{mj}w_m = \sum_{i=1}^m A_{ij}w_i.$$

Use these coordinate vectors as the columns of A :

$$A = \left[\begin{array}{c|ccc|c} & & & & \\ & | & & & | \\ & [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} & \\ & | & & & | \end{array} \right].$$

This constructs the required matrix. If $B = [B_{ij}]$ were another matrix with the same property, then for every j ,

$$\sum_{i=1}^m B_{ij}w_i = T(v_j) = \sum_{i=1}^m A_{ij}w_i.$$

The uniqueness of coordinates with respect to $\mathcal{C}$ gives $B_{ij} = A_{ij}$ for every i and j . Hence $B = A$.

Conversely, let $A = [A_{ij}]$ be an arbitrary $m \times n$ matrix and define

$$\tilde{w}_j = \sum_{i=1}^m A_{ij}w_i \quad (j = 1, \dots, n).$$

By the linear map lemma, there is a unique linear transformation $T: V \rightarrow W$ such that $T(v_j) = \tilde{w}_j$ for every j . Thus A determines a unique linear transformation with the required property. $\square$

If

$$v = x_1 v_1 + \cdots + x_n v_n,$$

then linearity gives

$$\begin{aligned} T(v) &= \sum_{j=1}^n x_j T(v_j) \\ &= \sum_{j=1}^n x_j \sum_{i=1}^m A_{ij} w_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n A_{ij} x_j \right) w_i. \end{aligned}$$

Consequently,

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Thus multiplication by the matrix A computes the coordinates of the image of a vector from the coordinates of the original vector.

6. SEPTEMBER 4, 2026

6.1. Matrix–vector multiplication. Let V and W be vector spaces over $\mathbb{F}$ with ordered bases

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m),$$

respectively, and let $T: V \rightarrow W$ be linear. Recall that the matrix representing T with respect to these bases is

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix},$$

where

$$T(v_j) = A_{1j} w_1 + \cdots + A_{mj} w_m.$$

Thus the j th column of A is the coordinate vector $[T(v_j)]_{\mathcal{C}}$.

Suppose that

$$v = x_1 v_1 + \cdots + x_n v_n, \quad [v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

By linearity,

$$\begin{aligned} [T(v)]_{\mathcal{C}} &= x_1 [T(v_1)]_{\mathcal{C}} + \cdots + x_n [T(v_n)]_{\mathcal{C}} \\ &= x_1 \begin{bmatrix} A_{11} \\ A_{21} \\ \vdots \\ A_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} A_{1n} \\ A_{2n} \\ \vdots \\ A_{mn} \end{bmatrix} \\ &= \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}. \end{aligned}$$

This motivates the following definition.

Definition 12 (Matrix–vector product). *Let $A = [A_{ij}] \in M_{m \times n}(\mathbb{F})$, and write its columns as $A_1, \dots, A_n$. For*

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n,$$

define

$$Ax = x_1 A_1 + \cdots + x_n A_n.$$

Equivalently, the k th coordinate of Ax is

$$(Ax)_k = \sum_{j=1}^n A_{kj} x_j, \quad k = 1, \dots, m.$$

With this notation, the coordinate formula for a linear transformation is

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Example 12. *Matrix–vector multiplication can be computed either as a linear combination of the columns or one coordinate at a time. For*

instance,

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 10 \end{bmatrix},$$

while

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 \\ 4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}.$$

6.2. Finding matrix representations.

Example 13. Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x - 5y \\ 7y \end{bmatrix}.$$

With respect to the standard bases, the columns of the representing matrix are

$$T(e_1) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad T(e_2) = \begin{bmatrix} 2 \\ -5 \\ 7 \end{bmatrix}.$$

Hence

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -5 \\ 0 & 7 \end{bmatrix}.$$

Example 14 (Differentiation). Let $n \geq 1$, and let

$$T: \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \quad T(p) = p',$$

and use the ordered basis

$$\mathcal{B} = (1, t, t^2, \dots, t^n)$$

for both the domain and codomain. Since

$$T(1) = 0, \quad T(t) = 1, \quad T(t^j) = jt^{j-1},$$

the matrix representing T is

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & n \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

6.3. The vector space of linear maps. Let V and W be vector spaces over $\mathbb{F}$. If $T: V \rightarrow W$ is linear and $\alpha \in \mathbb{F}$, define $\alpha T: V \rightarrow W$ pointwise by

$$(\alpha T)(v) = \alpha T(v).$$

This map is again linear. Indeed, for $u, v \in V$,

$$\begin{aligned} (\alpha T)(u + v) &= \alpha T(u + v) \\ &= \alpha(T(u) + T(v)) \\ &= (\alpha T)(u) + (\alpha T)(v), \end{aligned}$$

and, for $\beta \in \mathbb{F}$,

$$(\alpha T)(\beta u) = \alpha T(\beta u) = \alpha \beta T(u) = \beta(\alpha T)(u).$$

Similarly, if $T_1, T_2: V \rightarrow W$ are linear, define their sum by

$$(T_1 + T_2)(v) = T_1(v) + T_2(v).$$

The map $T_1 + T_2$ is linear because

$$\begin{aligned} (T_1 + T_2)(u + v) &= T_1(u + v) + T_2(u + v) \\ &= (T_1 + T_2)(u) + (T_1 + T_2)(v) \end{aligned}$$

and

$$(T_1 + T_2)(\alpha u) = \alpha(T_1 + T_2)(u).$$

Definition 13 (Space of linear maps). *The collection of all linear transformations from V to W is denoted by*

$$\mathcal{L}(V, W) = \{T: V \rightarrow W : T \text{ is linear}\}.$$

Proposition 9. *The set $\mathcal{L}(V, W)$, equipped with pointwise addition and scalar multiplication, is a vector space over $\mathbb{F}$.*

Proof. The preceding calculations show that the two operations are closed in $\mathcal{L}(V, W)$. All vector-space identities follow pointwise from the corresponding identities in W . The additive identity is the zero transformation $0: V \rightarrow W$, defined by $0(v) = 0$, and the additive inverse of T is the map $-T$, defined by $(-T)(v) = -T(v)$. $\square$

6.4. Composition of linear maps. Let V , W , and Z be vector spaces over $\mathbb{F}$. Suppose that

$$T_2: V \longrightarrow W \quad \text{and} \quad T_1: W \longrightarrow Z$$

are linear. Their *composition* is the map

$$T_1 \circ T_2: V \longrightarrow Z, \quad (T_1 \circ T_2)(v) = T_1(T_2(v)).$$

Proposition 10. *The composition $T_1 \circ T_2$ is linear.*

Proof. For $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{aligned} (T_1 \circ T_2)(u + v) &= T_1(T_2(u) + T_2(v)) \\ &= T_1(T_2(u)) + T_1(T_2(v)), \end{aligned}$$

and

$$(T_1 \circ T_2)(\alpha u) = T_1(\alpha T_2(u)) = \alpha T_1(T_2(u)).$$

$\square$